

Henry Lee

Lead Software Engineer · Backend Platforms & Distributed Systems

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8+ years	50k → 400k+	10M+ / day	1 → 30 days
SOFTWARE ENGINEERING	DAILY ACTIVE USERS	DATA RECORDS PROCESSED	QUERY RANGE

SUMMARY

Lead Software Engineer with 8+ years of experience building backend systems for a high-volume consumer transaction platform. Progressed from transaction integrations and reporting to building an in-house customer-support system, scaling services as the platform grew from **50k to 400k+ daily active users**, and designing recoverable workflows for user-facing transaction requests. Currently leads backend delivery while remaining hands-on with TypeScript, NestJS, gRPC, Temporal, MongoDB, and Redis.

EXPERIENCE

SNSoft Sdn Bhd

○ Lead Software Engineer (Backend Team Lead)

Dec 2025 – Present

- Lead backend development for the platform's transaction-request workflows and shared integration services, while continuing to design and implement production systems.
- Designed and coded reusable **Temporal workflows with idempotent activities** for requests that pass through multiple approval stages before execution, including deposits, withdrawals, and other monetary adjustments.
- Used durable state and automatic retries to eliminate recurring stuck requests during service restarts, external disconnections, and database load, removing the need for on-call engineers to run manual recovery scripts.
- Built a CDC and streaming-ingestion pipeline with **Apache Flink connectors**, including MongoDB Change Streams, to consolidate monetary-change records from 10+ upstream systems spanning MongoDB, Redis, ClickHouse, and other stores into a unified Hologres table processing over **10M records per day**.
- Developed the shared SDK and service contracts that allow multiple product teams to create transaction requests through a consistent interface.
- Introduced **Atlas Search** after MongoDB indexes reached their practical limit, supporting faster and fuzzy searches across transaction requests, deposits, withdrawals, and multi-platform reporting data.

○ Senior Software Engineer

Jan 2024 – Dec 2025

- Built an in-house customer-support chat system to replace a third-party product, using **NestJS, gRPC, and Matrix** to connect users with support agents.
- Added routing, agent queues, concurrency controls, AI-to-human escalation, and an agent view that brings together user details, transaction-request history, and KYC status during support conversations.
- Reduced MongoDB I/O pressure by moving high-frequency and short-lived data into **Redis, local memory, and TTL-based storage** as the platform grew from **50k to 400k+ daily active users**.
- Replaced unreliable WebSocket long-polling notifications with **Redis Pub/Sub** for real-time updates across service instances.

Software Engineer

Dec 2019 – Jan 2024

Implemented **multi-currency support** across deposit, withdrawal, and transfer workflows on the consumer transaction platform.

Built regulatory and operational reporting for datasets containing **millions of rows**, including automated file splitting to meet submission constraints and deadlines.

Integrated third-party providers and improved live-data connections with WebSocket heartbeats, reconnection, and failure handling.

Hokenso Sdn Bhd (Hitachi eBworx) — Mobile Engineer

Apr 2018 – Oct 2019

Built Great Eastern Singapore's first insurance **point-of-sale mobile application**, supporting agents in their daily sales workflow.

Developed responsive web and hybrid mobile features across delivery, bug fixing, testing support, and production deployment.

Owned QA deployments and **automated the daily release process** to improve speed and consistency.

EARLY CAREER

QA & Field-Test Engineer Intern, OPPO Electronics Malaysia HQ — field testing, troubleshooting, test-case design, and product-quality feedback (2017)

IT & Network Support Intern, Chin Hin — IT and network support across 23 branches (2014)

SKILLS

LANGUAGES	TypeScript (advanced) · JavaScript / Node.js 20+ · Protocol Buffers · SQL · Java and Rust (working)
BACKEND	NestJS · gRPC · Express · Socket.io / WebSocket · Temporal.io · Apache Pulsar
DATA	MongoDB / Mongoose (aggregation, sharding, Atlas Search) · PostgreSQL · Redis (Pub/Sub, Cluster) · ClickHouse · Hologres · Apache Flink (Java DataStream, Flink SQL) · change data capture (MongoDB to Hologres)
OBSERVABILITY & INFRA	Grafana · Aliyun SLS · OpenTelemetry · Prometheus · HashiCorp Vault · Docker · GitHub Actions · Alibaba Cloud · AWS S3
QUALITY	Jest · Testcontainers · JUnit 5 + JaCoCo · 80% coverage gate · weekly release train with full regression
AI ENGINEERING	Claude Code · Model Context Protocol (MCP) · artifact-gated agent pipelines · context engineering · NotebookLM module handbooks
PRACTICES	contract-first APIs (proto + JSON-Schema event registry) · event-driven architecture · idempotent and retry-safe processing · saga and compensation · domain-driven design · zero-downtime migrations

EDUCATION

Bachelor of Information Technology — HELP University · 2012–2017 · English · Mandarin · Malay